

WIARTON, ON, Canada

WMO: 716330

Lat: 44.748N

Lon: 81.111W

Elev: 222

StdP: 98.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.5	(c) -16.6	(d) -23.9	(e) 0.4	(f) -18.8	(g) -20.8	(h) 0.6	(i) -16.1	(j) 12.3	(k) -1.7	(l) 10.9	(m) -1.3	(n) 2.9	(o) 140	(p) 0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	28.6	21.9	26.9	20.9	25.5	19.9	22.9	27.0	21.9	25.6	21.0	24.2	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	16.6	25.1	20.6	15.7	24.1	19.7	14.8	23.2	68.7	27.2	65.1	25.6	61.7	24.2	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.2	(b) 9.0	(c) 8.1	(d) DB	(e) -24.0	(f) 31.4	(g) 4.7	(h) 1.4	(i) -27.4	(j) 32.3	(k) -30.2	(l) 33.1	(m) -32.9	(n) 33.9	(o) -36.3	(p) 34.9	(q) 27.0
(a) 10.2	(b) 9.0	(c) 8.1	(d) WB	(e) -24.3	(f) 24.8	(g) 4.7	(h) 0.8	(i) -27.6	(j) 25.4	(k) -30.4	(l) 25.9	(m) -33.0	(n) 26.4	(o) -36.4	(p) 27.0	(q) 27.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-6.1	-5.9	-1.5	5.0	11.4	16.4	19.2	18.7	15.3	9.4	3.2	-2.1
	DBStd	10.14	6.17	5.88	5.86	4.75	4.40	3.75	3.03	2.94	3.99	4.49	4.68	4.85
	HDD10.0	2153	498	446	358	162	34	1	0	0	4	68	208	374
	HDD18.3	4284	757	679	614	399	221	82	27	31	108	281	453	633
	CDD10.0	1054	0	0	3	13	77	193	284	269	162	48	4	1
	CDD18.3	143	0	0	0	0	5	24	52	43	17	2	0	0
	CDH23.3	928	0	0	1	4	55	173	339	246	102	7	0	0
	CDH26.7	158	0	0	0	0	8	34	58	43	15	0	0	0
Wind	WSAvg	3.7	4.5	4.3	3.9	4.0	3.3	2.7	2.8	2.9	3.2	4.0	4.3	4.4
Precipitation	PrecAvg	1049	104	70	81	69	72	74	74	91	109	89	110	127
	PrecMax	1289	206	154	167	117	179	130	205	197	200	166	247	221
	PrecMin	689	54	23	12	15	21	8	0	25	34	33	35	65
	PrecStd	150	37	34	32	26	36	30	52	47	38	33	42	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	9.9	18.9	23.3	27.6	29.7	30.1	30.3	28.7	24.3	17.5	12.4
		MCWB	8.9	8.5	14.0	15.9	20.3	22.7	22.9	22.3	21.8	18.8	13.5	11.3
	2%	DB	6.4	5.9	13.0	19.2	24.9	27.5	28.3	27.7	26.2	21.4	14.7	8.3
		MCWB	5.3	4.3	10.0	13.0	18.2	21.0	22.1	21.7	20.4	16.9	11.8	6.9
	5%	DB	3.8	3.5	9.2	15.9	21.8	25.2	26.7	25.9	24.1	19.0	12.5	6.0
		MCWB	2.7	2.1	6.3	10.6	16.1	19.6	21.0	20.4	19.4	15.7	10.2	4.8
	10%	DB	2.1	1.9	6.2	12.8	19.4	23.3	25.2	24.4	22.2	16.5	10.2	4.1
		MCWB	1.1	0.7	3.7	8.3	14.5	18.3	19.9	19.6	18.3	13.7	8.2	2.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	9.1	14.9	17.2	21.1	23.5	24.4	23.7	23.0	20.3	14.6	11.3
		MCDB	9.8	9.8	18.3	21.6	26.4	28.9	28.5	28.0	26.3	22.8	16.2	12.6
	2%	WB	5.6	4.6	10.4	14.2	19.0	21.8	23.0	22.4	21.5	17.7	12.4	7.2
		MCDB	6.2	5.8	12.3	18.7	23.3	26.3	26.9	26.4	24.7	20.3	14.3	8.0
	5%	WB	2.9	2.2	6.7	11.3	17.1	20.5	21.8	21.5	20.1	15.8	10.3	4.9
		MCDB	3.7	3.2	8.9	15.2	20.8	24.2	25.5	24.7	22.9	18.5	12.3	5.8
	10%	WB	1.2	0.8	3.9	8.8	15.0	19.2	20.8	20.5	18.9	14.0	8.4	2.9
		MCDB	2.0	1.8	6.0	12.3	18.5	22.4	24.1	23.3	21.6	16.1	10.1	4.0
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.8	7.9	9.1	10.3	10.0	9.5	9.4	9.6	7.8	6.5	5.7
		MCDBR	7.7	8.4	11.3	15.0	13.8	12.2	11.3	11.3	11.6	11.2	10.0	7.4
		MCWBR	7.4	7.6	8.7	10.1	8.4	7.2	6.4	6.4	6.8	7.7	8.2	6.9
	5% WB	MCDBR	7.4	8.0	10.8	13.9	12.5	10.8	9.3	9.3	10.1	9.9	9.4	7.3
		MCWBR	7.4	7.5	8.9	10.2	8.6	7.0	5.9	5.9	6.8	7.4	8.2	7.1
Clear-Sky Solar Irradiance	taub	0.285	0.300	0.306	0.351	0.372	0.401	0.404	0.399	0.367	0.350	0.316	0.283	
	taud	2.416	2.330	2.418	2.353	2.381	2.318	2.349	2.348	2.426	2.440	2.485	2.474	
	Ebn at Noon	838	886	933	911	897	869	861	852	854	819	795	808	
	Edh at Noon	72	95	100	117	117	125	120	116	99	85	69	63	
All-Sky Solar Radiation	RadAvg	1.12	2.02	3.40	4.56	5.52	6.05	6.11	5.20	4.01	2.34	1.28	0.81	
	RadStd	0.14	0.23	0.37	0.41	0.52	0.42	0.29	0.26	0.36	0.26	0.22	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page